# DAVID L. RUDLAND

**614-571-6355 (cell)**


## EDUCATION:

**Yokohama University,** Yokohama, Japan

Ph. D., Engineering, Division of Materials Science and Engineering, March 2005
*Dissertation:* "Predicting Cracked Pipe Response using Fracture Mechanics Techniques"
Applied current and past research through a visiting scholar program

**University of Illinois,** Urbana, Illinois

M.S., Mechanical Engineering, May 1993
Master's Thesis: "High Temperature Torsional Fatigue of Ceramics with Vitreous Grain Boundaries"
With honors

B.S., Mechanical Engineering, May 1988
With honors

## PROFESSIONAL INTERESTS:

- Material behavior
- Fracture mechanics of engineering materials
- Probabilistic fracture mechanics
- Piping design and analysis
- Welding simulation and integrity
- Small- and large-scale experimental testing
- Finite element analyses including stress analyses, failure prediction, and fracture mechanics
- Engineering design of structures

## RELEVANT EXPERIENCE:

**Rudland Consulting Services, LLC,** Frederick, Maryland – 2025-present
Owner
Providing engineering consulting services to the nuclear industry including:

- Fracture Mechanics, flaw evaluation, structural integrity, and aging management review and/or development for passive components
- Probabilistic fracture mechanics support for both long term operation for existing power plants and design support for ASME Section XI Division 2 RIM program development.
- Material selection, qualification, and deployment support for light water and non-light water reactor design.
- Regulatory advice and support through all stages of the licensing process for the operating and new reactor fleet.
- Training and mentoring for materials, structural integrity and aging behaviors for passive nuclear components.

**United States Nuclear Regulatory Commission,** Washington, DC – 2008-2025
Office of Nuclear Reactor Regulations, Division of New and Renewed Licenses
Senior Technical Advisor for Nuclear Power Plant Materials, 2017-2025
Duties include:

- Serving as a senior advisor and agency expert providing authoritative technical expertise, advice, and support on a broad variety of technical, regulatory, and policy issues related to the materials aging, structural integrity, inspection and performance aspects of nuclear power plant components, including reactor pressure vessel, reactor vessel internals, steam generators, piping and other pressure-retaining components.

- Providing authoritative technical advice and analysis for the materials aging, structural integrity, inspection and performance aspects of nuclear power plant components. Specific areas of technical review include aging mechanisms and management programs, deterministic and probabilistic structural integrity analyses, crack initiation and propagation, in-service inspection programs and techniques (e.g., eddy current, radiographic, ultrasonic, visual, magnetic particle and dye penetrant) and statistical analysis of data.
- Recommending modifications to NRC regulations, guides, and acceptance criteria based upon research results and the latest developments in mechanical/materials engineering, and counseling NRC management regarding the likely effects of technical, safety, regulatory, and programmatic actions.
- Coordinating crucial interactions and representing the office and agency with internal and external entities, including government, industry, and international nuclear regulatory organizations, including presenting and defending NRC technical positions, concerns, objectives, and requirements.
- Serving as a lead mentor for subordinate level staff. Provides guidance to staff members on technical analysis methods and regulatory approaches for resolving issues. Assists in developing overall staff analysis and evaluation capabilities in the areas of the materials aging, structural integrity, and materials inspection and performance issues aspects of reactor pressure vessels, reactor vessel internals, steam generators, piping, and other pressure retaining components in nuclear power plants.

Office of Nuclear Reactor Regulations, Division of Engineering
Chief, Vessel and Internals Integrity Branch, 2016 - 2017

- Lead and managed a group of 12 highly technical staff members with expertise in fracture mechanics, materials science, metallurgy, and mechanical engineering on
    - Integrity, aging management and regulatory issues concerning the reactor vessel and internals for currently operating light water reactors in the US
    - Integration of technical evaluations with probabilistic safety assessments to assess safety implications in a risk-informed decision-making manner
    - Identifying needs for new rulemaking and supporting research, providing support to guide research, drafting new and updated regulations and Regulatory Guides, and issuing generic communications as needed
    - Conducting safety evaluations of licensee's implementation of NRC requirements, changes to existing licenses (including license renewals), topical reports, relief requests and new licenses
    - Providing engineering expertise by assisting with inspections and projects.
    - Providing direct support to the NRC regions on highly technical and regulatory issues
- Represented the organization at national and international levels on both technical and regulatory topics.
- Represented the NRC at the ASME Board of Nuclear Codes and Standards, which manages of all ASME activities related to codes, standards and guides directly applicable to nuclear facilities and technology.
- Responsible for all of the administrative activities within the branch, e.g. coaching staff, set goals, monitor progress, establish training and career plans, reviewing and approving all products, and ensuring technical reviewers properly execute their projects to meet the needs of the agency.

Office of Nuclear Regulatory Research, Component Integrity Branch
Chief, Component Integrity Branch, 2013-2016

- Lead and managed group of 13 high-technical staff members with expertise in solid mechanics, fracture mechanics, stochastic modeling, non-destructive testing, materials science, metallurgy, and mechanical engineering with an annual external research budget of ~$7M focused on structural integrity issues of light water reactor primary coolant pressure boundary systems:
  - Reactor pressure vessel integrity (neutron embrittlement/pressurized thermal shock)
  - Environmental degradation of piping systems, control rod drive mechanisms, bottom mounted instrument penetrations, pressurizers, etc.
  - Polyethylene piping for use in Class 3 safety systems
  - Non-destructive testing to assess environmental degradation (UT, PT, ET, VT, etc.)
  - Probabilistic fracture mechanics code development
  - Numerical stress and fracture mechanics analysis
  - ASME Boiler & Pressure Vessel Code support
- Responsible for all of the technical and research activities and product lines associated with the branch.
- Ensures products meet regulatory needs and are well vetted and peer-reviewed in the technical community.
- Responsible for all of the administrative activities within the branch, e.g. coaching staff, set goals, monitor progress, establish training and career plans, establish the overall budget, and ensure project managers properly execute their project budgets to meet the needs of the agency.

Senior Materials Engineer, 2008-2013

- Plans, develops, and manages regulatory-based research programs pertaining to integrity issues with reactor coolant pressure boundary piping. Responsibilities include:
  - Coordination with the regulatory offices to ensure research meets regulatory need,
  - Support to regulatory and regional offices by conducting analyses and providing input and reviews for safety evaluations and special inspections,
  - Oversight of contractor and in-house analyses, including stress corrosion cracking evaluations, detailed finite element analyses, welding residual stress simulation and probabilistic fracture mechanics analyses, and
  - Program budget and contract management.
- Leads development of a modular-based probabilistic fracture mechanics computer code (xLPR) for the calculation of probability of failure for primary piping systems. Responsibilities include:
  - Facilitated the development of a cooperative program though an addendum to the Memorandum of Understanding between the Office of Nuclear Regulatory Research and the Electric Power Research Institute (EPRI),
  - Overall program coordination, documentation, communication and management of over 50 team members including NRC staff, NRC contractors and EPRI contractors.
  - Moderator for technical and personnel issues within the xLPR team,
  - Established regulatory needs, defined a project management and budget structure to include decision-making processes and decision documentation, coordinated with the appropriate NRC/EPRI project managers to ensure contracts were in place, identified resources for task groups, set and reviewed project budgets, schedules and milestones, facilitated issue resolution, established the project review structure, and oversaw the quality assurance process,
  - Ensured smooth progress of the project toward meeting both long- and short-term regulatory goals, and ensured communication between all task groups, and within all constituencies,
  - Ensured consistency of the various task groups' recommendations for code development, and facilitated code verification and validation,
  - Conducted successful feasibility/pilot study that was published in conferences, technical journals and summarized in a May 2012 NUREG report, and
  - Leading xLPR Version 2 development

- Represented the NRC on the Working Group on Pipe Flaw Evaluation, High Temperature Flaw Evaluation and the Working Group on Flaw Evaluation for the ASME Section XI code with responsibilities to ensure the code analyses, processes and procedures promote safety within the nuclear power industry.

**Engineering Mechanics Corporation of Columbus,** Columbus, Ohio
Research Leader and Company Principal, 1999-2008

- Developed, planned, managed and executed contract research for the U.S. Nuclear Regulatory Commission and private national and international oil and gas companies. Areas of research included:
  - Developed advanced computational fracture mechanics models for reactor coolant pressure piping, control rod drive mechanism nozzles, and many fracture mechanics-based specimens. Solutions developed were used to predict growth rates and probability of failure for stress corrosion cracking issues.
  - Developed probabilistic fracture mechanics code (PRO-LOCA) for predicting the frequency of occurrence of large-break loss of coolant accidents for the NRC and international industry.
  - Developed and implemented procedures for predicting welding residual stresses in butt welds using computational stress analysis for nuclear piping components.
  - Designed and conducted full- and small-scale fracture testing experiments for oil and gas, and nuclear grade piping in support of engineering analysis development.
- Mentored, supervised and appraised staff in both technical and career advancement areas.

# PROFESSIONAL AFFILIATIONS

**American Society of Mechanical Engineers**

- Member since 1999
- Elected Fellow in 2011
- NRC representative to the ASME Board of Nuclear Codes and Standards, and the Special Working group on Additive Manufacturing. Active on the Working Group on Pipe Flaw Evaluation, the Working Group on Flaw Evaluation, and the Working Group on Operating Plant Criteria for the ASME Section XI Committee
- Published numerous PVP and JPVT papers, see Publications
- Secretary to the Materials and Fabrication Committee within the Piping and Pressure Vessel Division (2010-2012).
- Chairman of the Materials and Fabrication committee within the Piping and Pressure Vessel Division (2012-2016)
- Associate Editor for the Journal of Pressure Vessel Technology (2012-2017)
- Session Developer, Session Chair and Co-Chair for PVP conferences in 2003-2024. Served as Technical Program Representative for Materials and Fabrication committee for PVP09.

**American Society for Testing and Materials**

- Member since 1994
- Active on committee E08.08, elastic plastic fracture and involved in standard development for fracture testing
- Published papers and acted as Session Chair in numerous ASTM symposium on Fracture (ASTM STP), see Publications

# AWARDS

- ASME Pressure Vessel and Piping Conference Award for the 2010 and 2011 PVP Jim Farr

Outstanding Conference Session
- ASME Pressure Vessel and Piping, S. S. Chen Outstanding Service Award in 2016.
- ASME S.Y Zamrik Pressure Vessel and Pipe Medal for outstanding contributions in the field of pressure vessel and piping technology including, but not limited to, research, development, teaching and significant advancement of the state-of-the-art in 2020.
- ASME Pressure Vessel and Piping Conference Award for Best Technical Paper of the 2023 PVP conference entitled, "A Review of Current Best Practice for Validation of Probabilistic Fracture Mechanics Codes for Assessment of Nuclear Structural Integrity."
- NRC Meritorious Service Award in recognition of his authoritative leadership and superb technical expertise in developing regulatory policy and solving technical problems in an efficient and risk-informed manner to ensure safe operation in 2024.

# SELECT PUBLICATIONS

**Technical Journal and Conference**

1. Rudland, D., Tregoning, R., and Homiack, M., "Determination of the Continued Adequacy of LOCA Frequencies From NUREG-1829," PVP2025-154873, Proceedings of the ASME 2025 Pressure Vessels & Piping Conference, PVP2025, July 20-25, 2025, Montreal, Quebec, Canada
2. Rudland, D., Leech, M., Homiack, M., Nellis, C., "Risk-Informed Assessment of French Stress Corrosion Cracking Operational Experience Relative to The Fleet of Pressurized Water Reactors in The United States of America," PVP2024-121559, Proceedings of the ASME 2024 Pressure Vessels & Piping Conference, PVP2024, July 29-August 2, 2024, Bellevue, Washington.
3. Hojo, K., Rudland, D., "Comparison Of K-Solutions of Rectangular Flaw with Semi-Elliptical Flaw for PWSCC Growth Analysis," ICONE31-123140, Proceedings of the 31st International Conference on Nuclear Engineering, ICONE31, August 4-8, 2024, Prague, Czech Republic.
4. Xu, S., Rudland, D., and Scarth, D., "A Review of Current Best Practice for Validation of Probabilistic Fracture Mechanics Codes for Assessment of Nuclear Structural Integrity," PVP2023-107513, Proceedings of the ASME 2023, Pressure Vessels & Piping Conference, PVP2023, July 16-21, 2023, Atlanta, GA, USA
5. Rudland, D., and Widrevitz, D., "Statistical Approach to Developing a Performance Monitoring Program," PVP2023-105203, Proceedings of the ASME 2023, Pressure Vessels & Piping Conference, PVP2023, July 16-21, 2023, Atlanta, GA, USA
6. Rudland, D., Cumblidge, S., Ng, C., and Anzalone, R., "Predicting the Impact of CRDM Thermal Sleeve Flange Wear in Westinghouse Pressurized Water Reactors," Journal of Pressure Vessel Technology, Transactions of the ASME, June 2022, Vol. 144 / 031501
7. Rudland, D., "Can Inservice Inspections Be Eliminated for Passive Components in the U.S. Nuclear Fleet?," PVP2022-78318, Proceedings of the ASME 2022, Pressure Vessels & Piping Conference, PVP2022, July 17-22, 2022, Las Vegas, Nevada, USA
8. Rudland, D., "Predicting Pipe Rupture Frequencies Using xLPR," PVP2020-21080, Proceedings of the ASME 2020 Pressure Vessels & Piping Division Conference, PVP2020, July 19 – 24, 2020, Minneapolis, Minnesota.
9. Rudland, D., Cumblidge, S., Ching, NG, and Anzalone, R., "Predicting the Impact of CRDM Thermal Sleeve Wear in Westinghouse Pressurized Water Reactors," SMiRT-25, Division II, Charlotte, NC, USA, August 4-9, 2019.
10. Lyons, S., Homiack, M., Rudland, D." Applying Risk-Informed Decision-Making to Evaluate the Safety Significance of Potential Piping Failures Due to Primary Water Stress Corrosion Cracking, SMiRT-25, Division II, Charlotte, NC, USA, August 4-9, 2019.
11. Kalyanam, S., Brust, F., Shim, D.J., Wilkowski, G., and Rudland D., "Validation of Axial Crack Stability Assessment Methods in XLPR Version 2.0 Code," PVP2016-63830, Proceedings of the ASME 2016 Pressure Vessels & Piping Conference, PVP2016, July 17-21, 2016, Vancouver, BC Canada

12. Rudland, D.L., Kirk, M., and Raynaud, P., "Thoughts on Improving Confidence in Probabilistic Fracture Mechanics Analyses," PVP2016-63771, Proceedings of the ASME 2016 Pressure Vessels & Piping Conference, PVP2016, July 17-21, 2016, Vancouver, BC Canada
13. Uddin, M., Wilkowski, G., Brust, F., Shim, D.J., and Rudland, D.L., "Validation of Axial Crack Stability Assessment Methods In xLPR Version 2.0 Code," PVP2016-63830, Proceedings of the ASME 2016 Pressure Vessels & Piping Conference, PVP2016, July 17-21, 2016, Vancouver, BC Canada
14. Shim, D.J., and Rudland, D.L., "Deterministic Analysis Results Supporting xLPR Version 2 Code Development," PVP2016-63055, Proceedings of the ASME 2016 Pressure Vessels & Piping Conference, PVP2016, July 17-21, 2016, Vancouver, BC Canada
15. Rudland, D., Harrington, C., and Dingreville, R. "Development of The Extremely Low Probability of Rupture (xLPR) Version 2.0 Code," PVP2015-45134, Proceedings of the ASME 2015 Pressure Vessels & Piping Conference, PVP2015, July 19-23, 2015, Boston, MA
16. Rudland, D.L., Benson, M., Shim, D.-J., "Fracture Toughness Behavior of Complex Cracks in Dissimilar Metal Welds," PVP2014-28105, Proceedings of the ASME 2014 Pressure Vessels & Piping Conference, PVP2014, July 20-24, 2014, Anaheim, CA
17. Shim, D.-J., Rudland, D.L., and Park, J.-S., "Surface to Through-Wall Crack Transition Model for Axial Cracks in Pipes," PVP2014-28048, Proceedings of the ASME 2014 Pressure Vessels & Piping Conference, PVP2014, July 20-24, 2014, Anaheim, CA
18. DeWald, A.T., Hill, M, Benson, M., and Rudland, D.L., "Measurement of Residual Stress in Reactor Mockups Containing Dissimilar Metal Welds," PVP2014-28209, Proceedings of the ASME 2014 Pressure Vessels & Piping Conference, PVP2014, July 20-24, 2014, Anaheim, CA
19. Rudland, D.L., Lukes, R., and Shim, D.-J., "Fracture Toughness Behavior of Throughwall Cracks in Dissimilar Metal Welds," PVP2013-97143, Proceedings of the ASME 2013 Pressure Vessels & Piping Conference, PVP2013, July 14-18, 2013, Paris, France
20. Brust, F.W., Rudland, D.L., Rathbun, H., and Shim, D.-J., "Weld Residual Stress Analysis and Axial PWSCC Predictions in A Double Vee Groove Large Diameter Nozzle," PVP2013-98065, Proceedings of the ASME 2013 Pressure Vessels & Piping Conference, PVP2013, July 14-18, 2013, Paris, France
21. Zhu, X-K., Shim, D.-J., Rudland, D.L., and Scott, P., "Determination Of J-R Curves for Dissimilar Metal Welds Using Normalization Method and Compact Tension Specimens," PVP2013-98051, Proceedings of the ASME 2013 Pressure Vessels & Piping Conference, PVP2013, July 14-18, 2013, Paris, France
22. Shim, D.-J., Kurth, B., and Rudland, D.L., "Development of Non-Idealized Surface To Through-Wall Crack Transition Model," PVP2013-97092, Proceedings of the ASME 2013 Pressure Vessels & Piping Conference, PVP2013, July 14-18, 2013, Paris, France
23. Shim, D-J., Rudland, D.L., Brust, F.W., "Comparison of Through-Wall and Complex Crack Behaviors in Dissimilar Metal Weld Pipe Using Cohesive Zone Modeling", PVP2013-98041, Proceedings of the ASME 2013 Pressure Vessels & Piping Conference, PVP2013, July 14-18, 2013, Paris, France
24. Rudland, D. L., Shim D-J., and Xu S.," Simulating Natural Axial Crack Growth In Dissimilar Metal Welds Due To Primary Water Stress Corrosion Cracking," PVP2013-97188, Proceedings of the ASME 2013 Pressure Vessels & Piping Conference, PVP2013, July 14-18, 2013, Paris France
25. Rudland, D., and Harrington, C., "Initial Development of the Extremely Low Probability of Rupture (xLPR) Version 2.0 Code," PVP2012-78186, Proceedings of the ASME 2012 Pressure Vessels & Piping Conference, PVP2012, July 15-19, 2012, Toronto, ON, Canada.
26. Rudland, D., Scott, P., Olson, R., Cox, A., Shim, D-J. and Lukes, R., "Dissimilar Metal Weld Pipe Fracture Testing– Analysis of Results and their Implications," PVP2012-78140, Proceedings of the ASME 2012 Pressure Vessels & Piping Conference, PVP2012, July 15-19, 2012, Toronto, ON, Canada.
27. Cox, A., Scott, P., Olson, R., Young, B., and Rudland, D., "Dissimilar Metal Weld Pipe Fracture

Testing – Experimental Procedures and Results," PVP2012-78353, Proceedings of the ASME 2012 Pressure Vessels & Piping Conference, PVP2012, July 15-19, 2012, Toronto, ON, Canada

28. Benson, M., Rudland, D., and Kirk, M., "Thermomechanical Analysis of Pressure Vessels," PVP2012-78229, Proceedings of the ASME 2012 Pressure Vessels & Piping Conference, PVP2012, July 15-19, 2012, Toronto, ON, Canada
29. Do-Jun Shim, Gery M. Wilkowski, and David L. Rudland, "Determination of the Elastic-plastic Fracture Mechanics Z-factor for Alloy 182 Weld Metal Flaws," *International Journal of Pressure Vessels and Piping*, 88 (2011), 231-238.
30. Rudland, D., and Harrington C., Extremely Low Probability of Rupture (xLPR) Version 1.0 Code – Pilot Study Problem Results," PVP2011-57531. Proceedings of the ASME 2011 Pressure Vessels & Piping Conference, PVP2011, July 17-21, 2011, Baltimore, Maryland, USA.
31. Rudland, D., Scott, P., Olson, R., and Cox, A., "Complex Crack Stability in Dissimilar Metal Welds, Background and Test Plan," PVP2011-57535. Proceedings of the ASME 2011 Pressure Vessels & Piping Conference, PVP2011, July 17-21, 2011, Baltimore, Maryland, USA.
32. Rudland, D., Brust, F., Shim, DJ., and Wilkowski, G., "PWSCC Crack Growth Mitigation With Inlay," PVP2011-57954. Proceedings of the ASME 2011 Pressure Vessels & Piping Conference, PVP2011, July 17-21, 2011, Baltimore, Maryland, USA.
33. Brust, F., Shim, D-J., Zhang, T., Wilkowski, G., and Rudland, "Modeling Crack Growth In Weld Residual Stress Fields Using The Finite Element Alternating Method," PVP2011-57935. Proceedings of the ASME 2011 Pressure Vessels & Piping Conference, PVP2011, July 17-21, 2011, Baltimore, Maryland, USA.
34. Shim, D-J., Harris, D., and Rudland, D., "Modeling of Subcritical Crack Growth Due to Stress Corrosion Cracking - Transition From Surface Crack to Through-Wall Crack," PVP2011-57267. Proceedings of the ASME 2011 Pressure Vessels & Piping Conference, PVP2011, July 17-21, 2011, Baltimore, Maryland, USA.
35. Rathbun, H., Fredette, L., and Rudland D., "NRC Welding Residual Stress Validation Program International Round Robin Program and Findings," PVP2011-57642. Proceedings of the ASME 2011 Pressure Vessels & Piping Conference, PVP2011, July 17-21, 2011, Baltimore, Maryland, USA.
36. Brust, F., Shim, DJ., Wilkowski, G., and Rudland D., "PWSCC Crack Growth Modeling Approaches," PVP2011-57974. Proceedings of the ASME 2011 Pressure Vessels & Piping Conference, PVP2011, July 17-21, 2011, Baltimore, Maryland, USA.
37. Brust, F., Kurth, R., Shim, DJ., and Rudland D., "Strategies for Treating Weld Residual Stresses in Probabilistic Fracture Mechanics Codes," PVP2011-57937. Proceedings of the ASME 2011 Pressure Vessels & Piping Conference, PVP2011, July 17-21, 2011, Baltimore, Maryland, USA.
38. Kerr, M., Rudland, D., Prime, M., Swenson, H., Buechler, M., "Characterization of a Plate Specimen from Phase I of the NRC/EPRI Weld Residual Stress Program," PVP2011-57687. Proceedings of the ASME 2011 Pressure Vessels & Piping Conference, PVP2011, July 17-21, 2011, Baltimore, Maryland, USA.
39. D. Rudland, T. Zhang, G. Wilkowski, and A. Csontos, "Welding Residual Stress Solutions for Dissimilar Metal Surge Line Nozzles Welds," *Journal of Pressure Vessel Technology*, April 2010, Vol. 132.
40. D. Rudland, D.-J. Shim and A. Csontos, "Natural Flaw Shape Development Due To Stress Corrosion Cracking," *Journal of Pressure Vessel Technology*, February 2010, Vol. 132
41. Rudland, D., Brust, F., Shim, DJ, Punch, E.F., and Kalyanam, S., "Natural PWSCC Crack Growth in Dissimilar Metal Welds With Inlay," EPRI 2010 International Pressurized Water Reactor Conference and Exhibition, Colorado Springs, Colorado, June 30, 2010
42. Brust, F., Rudland, D., Zhang, T., Further Aspects of Three Dimensional Computational Weld Modeling, EPRI 2010 International Pressurized Water Reactor Conference and Exhibition, Colorado Springs, Colorado, June 30, 2010.
43. Rudland, D., Harrington, C., "Alloy 600 Management - Current and Future Plans, xLPR Modular Code," EPRI 2010 International Pressurized Water Reactor Conference and Exhibition, Colorado

Springs, Colorado, June 30, 2010.

44. Rudland, D., Brust, F.,Zhang, T., “Further Welding Residual Stress And Flaw Tolerance Assessment Of Dissimilar Metal Welds With Alloy 52 Inlays, PVP2010-25433, Proceedings of the ASME 2010 Pressure Vessels & Piping Division / K-PVP Conference, PVP2010, July 18-22, 2010, Bellevue, Washington, USA.
45. Rudland, D., Mattie, P., Klasky, H., Harris, D., Bishop, B., and Kurth, R., “Development Of Computational Framework And Architecture For Extremely Low Probability Of Rupture (xLPR) Code,” PVP2010-25963, Proceedings of the ASME 2010 Pressure Vessels & Piping Division / K-PVP Conference, PVP2010, July 18-22, 2010, Bellevue, Washington, USA.
46. Brust, F.,., Shim, DJ, Punch, E.F., Kalyanam S., and Rudland, D. “Natural PWSCC Crack Growth In Dissimilar Metal Welds With Inlay,” PVP2010-26108. Proceedings of the ASME 2010 Pressure Vessels & Piping Division / K-PVP Conference, PVP2010, July 18-22, 2010, Bellevue, Washington, USA.
47. Rudland, D., Csontos, A, Brust, F., and Zhang, T., “Continuing Development of Welding Residual Stress And Flaw Evaluation For Dissimilar Metal Welds With Alloy 52 Inlays,” PVP2009-77167, Proceedings of ASME-PVP 2009: 2009 ASME Pressure Vessels and Piping Division Conference, July 26-30, 2009, Prague, Czech Republic.
48. Rudland, D., Scott, P., Kurth, R., and Cox A., “Continuing Development of PRO-LOCA for the Prediction Of Break Probabilities For Loss-Of-Coolant Accidents,” PVP2009-77053, Proceedings of ASME-PVP 2009: 2009 ASME Pressure Vessels and Piping Division Conference, July 26-30, 2009, Prague, Czech Republic.
49. Zhang, T., Brust, F., Wilkowski, G., Rudland, D., and Csontos A., “Welding Residual Stress and Multiple Flaw Evaluation for Reactor Pressure Vessel Head Replacement Welds with Alloy 52,” PVP2009-78112, Proceedings of ASME-PVP 2009: 2009 ASME Pressure Vessels and Piping Division Conference, July 26-30, 2009, Prague, Czech Republic.
50. Shim, D.-J., Kurth, E., Brust, F., Wilkowski, G., Rudland, D., and Csontos A., “Crack-Opening Displacement And Leak-Rate Calculations For Full Structural Weld Overlays,” PVP2009-77966, Proceedings of ASME-PVP 2009: 2009 ASME Pressure Vessels and Piping Division Conference, July 26-30, 2009, Prague, Czech Republic.
51. D. Rudland, D.-J. Shim and A. Csontos, “Natural Flaw Shape Development Due To Stress Corrosion Cracking,” PVP2008-61205, Proceedings of ASME-PVP 2008, 2008 ASME Pressure Vessels and Piping Division Conference, July 27-31, 2008, Chicago, IL, USA.
52. D.-J. Shim, D. Rudland, G. Wilkowski, B. Brust, and K. Ogawa, “Development Of Z-Factor For Circumferential Part-Through Surface Cracks In Dissimilar Metal Welds,” PVP2008-61110, Proceedings of ASME-PVP 2008, 2008 ASME Pressure Vessels and Piping Division Conference, July 27-31, 2008, Chicago, IL, USA.
53. F. Brust and D. Rudland, “Three Dimensional Aspects Of Computational Weld Modeling, PVP2008-61558, Proceedings of ASME-PVP 2008, 2008 ASME Pressure Vessels and Piping Division Conference, July 27-31, 2008, Chicago, IL, USA.
54. W. Bamford, R. Cipolla, D. Rudland, and G. Deboo, “Technical Basis for Revisions to IWB 3640 for Alloy 600/82/182 Flaw Evaluation in both PWR and BWR Environments,” PVP2008-61840, Proceedings of ASME-PVP 2008, 2008 ASME Pressure Vessels and Piping Division Conference, July 27-31, 2008, Chicago, IL, USA.
55. T. Zhang, G. Wilkowski, D. Rudland, F. Brust, H. Mehta, D. Sommerville, and Y. Chen, “Weld-Overlay Analyses – An Investigation of the Effect of Weld Sequencing,” PVP2008-61560, Proceedings of ASME-PVP 2008, 2008 ASME Pressure Vessels and Piping Division Conference, July 27-31, 2008, Chicago, IL, USA.
56. D. Rudland, Y. Chen, Wilkowski, G., and Csontos, A., “Effect of Hardening Law Assumption on weld Residual Stress Predictions,” 2007 PWSCC of Alloy 600 International Conference & Exhibit Show, Renaissance Waverly Hotel, Atlanta, GA, June 11-14, 2007

57. D. Rudland, Shim, D.-J., Xu, H., Wilkowski, G., and Csontos, A., "Evaluation of Wolf Creek Pressurizer Flaw Indications," 2007 PWSCC of Alloy 600 International Conference & Exhibit Show, Renaissance Waverly Hotel, Atlanta, GA, June 11-14, 2007
58. A. Csontos, D. Rudland, D.-J. Shim, H. Xu, and G. Wilkowski, "Flaw Evaluation of Wolf Creek Pressurizer Nozzle Indications," 13th International Conference on Environmental Degradation in Nuclear Power Systems, Whistler, British Columbia, August 19 - 23, 2007
59. Scott, P., Brust, F., Rudland, D., and Wilkowski, G., "Overview of the MERIT (Maximizing Enhancements in Risk-Informed Technology) Program," *Transactions*, SMiRT 19, Toronto, August 2007.
60. Rudland, D.L., Chen, Y. Zhang, T., Wilkowski, G., Broussard, J., and White, G., "Comparison of Welding Residual Stress Solutions for Control Rod Drive Mechanism Nozzles," PVP2007-26045, Proceedings of ASME-PVP 2007:, 2007 ASME Pressure Vessels and Piping Division Conference, July 22-26, 2007, San Antonio, TX, USA
61. Wilkowski, G., Rudland, D.L., Chen, Y., and Zhang, T., "Determination of the Elastic-Plastic Fracture Mechanics Z-factor for Alloy 82/182 Weld Metal Flaws for Use in the ASME Section XI Appendix C Flaw Evaluation Procedures," PVP2007-26733, Proceedings of ASME-PVP 2007:, 2007 ASME Pressure Vessels and Piping Division Conference, July 22-26, 2007, San Antonio, TX, USA
62. Simonen, F., Gosselin, S., Lydell, B., Wilkowski, G., and Rudland, D.L., "Application Of Failure Event Data To Benchmark Probabilistic Fracture Mechanics Computer Codes" PVP2007-26373, Proceedings of ASME-PVP 2007:, 2007 ASME Pressure Vessels and Piping Division Conference, July 22-26, 2007, San Antonio, TX, USA
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